Detecting Citation Drift in AI-Generated Literature Summaries

A Review of Concepts, Detection Methods, and Open Research Problems

Abstract

Large language models (LLMs) are increasingly used to summarize, synthesize, and cite scientific literature, yet the reliability of the citations embedded in their outputs remains poorly understood. This paper examines citation drift, a phenomenon recently formalized in the literature as the mutation, disappearance, or fabrication of references across repeated or extended interactions with an LLM (Ram, 2025). We situate citation drift within the broader hallucination literature (Ji et al., 2023; Huang et al., 2025), distinguishing it from single-shot citation fabrication (Walters & Wilder, 2023; Mugaanyi et al., 2024) and from the generalization bias that affects the semantic content of AI-generated summaries independently of their reference lists (Peters & Chin-Yee, 2025). Drawing on empirical studies of citation accuracy, hallucination detection, and retrieval-augmented generation, we synthesize a working taxonomy of citation drift mechanisms, review current detection approaches, including natural language inference (NLI)-based verification (Kryscinski et al., 2020), sampling-based consistency checks (Manakul et al., 2023), and retrieval-grounded citation verifiers (Khajavi et al., 2026), and assess their applicability to the specific problem of reference instability in literature summarization. We then identify methodological, evaluative, and infrastructural gaps that constrain progress, including the absence of standardized drift benchmarks, the conflation of factual and referential hallucination, and the limited coverage of non-English and low-resource scholarly corpora. We conclude by outlining a research agenda centered on longitudinal, multi-turn evaluation protocols, hybrid retrieval-verification pipelines, and shared metrics for reporting citation reliability in AI-assisted scholarship.

Keywords: *citation drift; hallucination detection; large language models; literature summarization; retrieval-augmented generation; citation fabrication; natural language inference; factual consistency; scholarly communication; AI-assisted research*

1. Introduction

The integration of large language models into scientific workflows has accelerated markedly since 2022, with researchers, students, and practitioners routinely using conversational AI systems to summarize papers, draft literature reviews, and generate bibliographies (Zhang & Zhao, 2025). This shift promises considerable efficiency gains: an LLM can condense a dense empirical article into a readable paragraph in seconds, and can, in principle, attach citations that allow a reader to trace claims back to their sources. In practice, however, the citations produced by these systems are frequently unreliable. Early audits of ChatGPT-generated bibliographies found that more than half of the references produced by GPT-3.5 did not correspond to any real publication, a rate that fell to still-substantial levels with GPT-4 (Walters & Wilder, 2023). Subsequent cross-disciplinary and cross-platform studies have confirmed that citation fabrication, while diminishing as models improve, remains a persistent feature of LLM-assisted writing across the natural sciences, humanities, and medicine (Mugaanyi et al., 2024; Bhattacharyya et al., 2023; Zhang & Zhao, 2025).

Most of this literature has treated citation unreliability as a static, single-response phenomenon: a model is prompted once, and the resulting references are checked against bibliographic databases. Yet contemporary use of LLMs for literature summarization is rarely a single exchange. Researchers iteratively refine prompts, ask follow-up questions, request expansions or condensations of the same summary, and revisit sources across multi-turn conversations. Ram (2025) recently introduced the term citation drift to describe what happens to references under these conditions: the same underlying claim, when re-summarized or re-cited across turns, can accrue a mutated author list, lose its citation entirely, or acquire a citation to a work that does not exist. Because citation drift concerns the stability of references over the course of an interaction rather than their accuracy in a single response, it constitutes a distinct evaluation problem that existing single-shot fabrication studies do not directly address.

This paper reviews the state of research on detecting citation drift in AI-generated literature summaries. We make three contributions. First, we synthesize a taxonomy that distinguishes citation drift from adjacent phenomena, namely single-shot citation fabrication, factual hallucination in summary content, and the broader class of generalization bias documented in AI science communication (Peters & Chin-Yee, 2025). Second, we review the technical machinery available for detecting reference instability, spanning natural language inference and question-answering based faithfulness metrics developed for abstractive summarization (Kryscinski et al., 2020; Ji et al., 2023), sampling-based hallucination detectors (Manakul et al., 2023), and retrieval-grounded citation verification systems purpose-built for scholarly text (Khajavi et al., 2026). Third, we identify concrete research gaps, including the absence of standardized multi-turn drift benchmarks, limited attention to non-English scholarship, and the difficulty of disentangling model-side drift from retrieval-side volatility in systems that combine LLMs with live web search. The remainder of the paper is organized as follows. Section 2 reviews related work on citation hallucination, summarization faithfulness, and retrieval-augmented generation. Section 3 presents a technical discussion of citation drift, including its proposed taxonomy and measurement. Section 4 surveys current detection approaches and benchmarks. Section 5 discusses limitations of the existing literature. Section 6 outlines research gaps and future directions, and Section 7 concludes.

2. Background and Related Work

2.1 Citation Fabrication in Single-Response Settings

The earliest and most extensively documented form of citation unreliability in LLM output is outright fabrication: a reference that is formatted correctly but does not correspond to any real publication. Walters and Wilder (2023) prompted GPT-3.5 and GPT-4 to produce 84 short literature reviews across 42 multidisciplinary topics and manually verified all 636 resulting citations against library databases; 55% of GPT-3.5 citations and 18% of GPT-4 citations were entirely fabricated, and among the non-fabricated citations, 43% (GPT-3.5) and 24% (GPT-4) contained substantive errors in author names, dates, journal titles, or page numbers. Comparable rates have been reported in biomedical contexts: Bhattacharyya et al. (2023) found that 47% of references generated by ChatGPT for medical case reports were fabricated and only 7% were both authentic and fully accurate, with incorrect PMID numbers appearing in 93% of the papers examined. Mugaanyi et al. (2024) extended this line of work cross-disciplinarily, finding that roughly a quarter of ChatGPT-generated citations across the natural sciences and humanities could not be verified, with substantial disparities in the presence of digital object identifiers (DOIs) between fields. More recent replications suggest gradual improvement but persistent unreliability: an eight-chatbot comparison found that six of eight systems fabricated a substantial share of requested bibliographic references, with fabrication rates ranging from near zero to well over one-third depending on the platform (arXiv:2505.18059).

Clinical commentary on these findings has emphasized that fabricated citations are not merely technical curiosities but a threat to the integrity of the biomedical literature, since plausible-looking but nonexistent references can be inadvertently propagated by authors who do not verify them before submission (Zhang & Zhao, 2025). Editorial responses have called for a combination of retrieval-augmented generation, publisher-specific fine-tuned models, and mandatory human verification as safeguards (Zhang & Zhao, 2025).

2.2 Hallucination in Abstractive Summarization

A parallel and longer-standing literature addresses hallucination in the content of machine-generated summaries, independent of citation behavior. Ji et al. (2023) provide a comprehensive taxonomy distinguishing intrinsic hallucination, in which the generated text contradicts the source, from extrinsic hallucination, in which the generated text cannot be verified against the source at all. Detection methods developed for this problem fall broadly into fact-based, classifier-based, question-answering-based, uncertainty-based, and LLM-based families (Huang et al., 2025). FactCC, a weakly supervised classifier trained on rule-based sentence transformations, was among the first models to jointly predict factual consistency and localize the supporting or contradicting span within the source document (Kryscinski et al., 2020). Question-answering-based metrics operate by generating questions from a candidate summary and checking whether the same answers are recoverable from the source text, while natural language inference (NLI)-based metrics treat the source as a premise and the summary as a hypothesis to be entailed (Huang et al., 2025). More recent sampling-based approaches, exemplified by SelfCheckGPT, dispense with an external reference altogether: because a model that truly 'knows' a fact tends to reproduce it consistently across repeated stochastic samples, while a hallucinated fact tends to vary, inter-sample agreement itself becomes a usable signal of unreliability (Manakul et al., 2023).

This body of work is directly relevant to citation drift because the underlying detection primitives, entailment checking, question-answering consistency, and sampling-based agreement, are the same primitives that recent citation-specific tools repurpose for reference verification (see Section 4). However, standard summarization-faithfulness metrics were designed to evaluate whether the prose of a summary is supported by a source document, not whether a structured bibliographic citation attached to that prose correctly identifies a real, matching publication. Citation drift research must therefore adapt these tools to a different object of evaluation: the citation as a discrete, checkable metadata record rather than a free-text claim.

2.3 Generalization Bias and Content-Level Fidelity

A distinct but related concern is generalization bias, in which an LLM-generated summary preserves accurate citations but overstates the scope of the findings being cited. Peters and Chin-Yee (2025) tested ten prominent LLMs, including GPT-4o, GPT-4.5, DeepSeek, LLaMA 3.3 70B, and Claude 3.7 Sonnet, on 4,900 summaries of scientific texts and found that most models produced broader generalizations than the original studies supported, even when explicitly prompted for accuracy; LLM-generated summaries were nearly five times more likely to contain broad generalizations than human-authored summaries (odds ratio = 4.85, 95% CI [3.06, 7.70]). Notably, they found that newer, larger models did not necessarily generalize more accurately than older ones, and that Claude's outputs remained closest to the original text in generalization scope among the models tested. This finding matters for citation drift research because it illustrates that reference-level fidelity (does the citation exist and match the right work) and content-level fidelity (does the summary accurately represent what the cited work found) are separable failure modes; a technically correct citation can still anchor a distorted claim, and drift-detection systems that check only for the existence of a reference will not catch this class of error.

2.4 Retrieval-Augmented Generation and the Citation Bias of Parametric Recall

Retrieval-augmented generation (RAG) was proposed as a general remedy for hallucination in knowledge-intensive NLP tasks by conditioning generation on retrieved passages from a non-parametric memory rather than relying solely on facts encoded in model weights (Lewis et al., 2020). Applied to citation generation, RAG-style pipelines such as ScholarCopilot dynamically decide when to retrieve literature and condition generation on the retrieved references, substantially improving retrieval accuracy relative to purely parametric baselines (Wang et al., 2025). Yet even when LLMs draw on genuine bibliometric knowledge rather than fabricating references outright, systematic biases persist in which references are selected. Analyzing 274,951 references generated by GPT-4o across 10,000 papers, Algaba et al. (2025) found that LLM-recommended citations systematically favor more recent work, shorter titles, and fewer authors than human citation practice, and that these systems reinforce the Matthew effect by disproportionately citing already highly-cited papers. A related study using GPT-4, GPT-4o, and Claude 3.5 in a controlled, non-retrieval setting found that models reproduce human citation biases in an amplified form even when generating suggestions purely from parametric knowledge (Algaba et al., 2024). These findings are important context for citation drift: even a model whose citations are individually verifiable may be systematically reshaping which literature gets surfaced and re-surfaced across a conversation, which is itself a subtler and more consequential form of drift than outright fabrication.

3. Citation Drift: Definition, Taxonomy, and Mechanisms

3.1 Defining Citation Drift

Ram (2025) defines citation drift as changes in a model's cited references, through mutation, loss, or fabrication, when the model responds to semantically equivalent prompts or continues an existing conversation. This definition deliberately extends beyond the single-response fabrication rate reported in the studies reviewed in Section 2.1, because it captures reference behavior as a trajectory across turns rather than as a property of one isolated output. In an accompanying formulation, citation drift has been framed as a failure of what the same research calls referential reproducibility, one of three layers of reproducibility that scientific LLMs should in principle satisfy alongside textual reproducibility (producing identical text under fixed decoding settings) and epistemic reproducibility (maintaining stable reasoning chains over time). Citation drift specifically targets the second layer: whether a model can reliably reproduce the same, correct citation for the same underlying claim when asked again, rephrased, or extended across a conversation.

In an empirical study spanning 240 conversations across four LLaMA models and 36 authentic scientific papers drawn from six domains, this research documented substantial citation instability and proposed two novel metrics for quantifying it: citation drift entropy, which measures the dispersion of cited references for semantically equivalent queries, and willingness-to-cite, which captures how consistently a model attaches any citation at all rather than omitting attribution altogether (Ram, 2025). Stability was found to vary markedly across models, underscoring that citation drift is not a uniform property of 'LLMs' in general but a measurable, model-specific behavior that can be benchmarked.

3.2 A Working Taxonomy of Drift Mechanisms

Building on this foundational work and the broader hallucination literature, we distinguish four mechanisms by which citations can drift across an AI-generated summarization workflow.

- Citation mutation: the cited work's identity is altered but not entirely lost, for example an author's name is swapped, a publication year shifts, a volume or page number changes, or an actual paper is cited but attributed with content it does not contain. Mutation overlaps substantially with the 'accurate but erroneous' citation category documented by Walters and Wilder (2023) and Mugaanyi et al. (2024), but the drift framing emphasizes that the same citation can mutate differently across repeated queries about the same claim.
- Citation loss: a claim that was previously attributed to a specific source in one turn is repeated later in the conversation without any citation, or with a vaguer, unattributable gesture toward 'prior research'. Loss is difficult to detect with single-response auditing because it requires comparing citation presence across turns rather than checking any one output in isolation.
- Citation fabrication: an entirely nonexistent reference is introduced, consistent with the single-shot fabrication behavior documented extensively in Section 2.1, but here considered as it recurs, changes, or newly appears across a multi-turn exchange rather than as a static rate.
- Referential substitution under retrieval volatility: in systems that combine an LLM with live web or database retrieval, the set of documents available to the model can itself change between calls due to index updates, ranking changes, or query reformulation, producing citation instability that originates in the retrieval layer rather than in the language model's parametric behavior. This mechanism is conceptually distinct from the other three because it can occur even when the underlying language model is fully deterministic, and it has been most visibly documented in commercial answer-engine contexts, where the sources underlying a generated answer are reported to rotate substantially from one query to the next even for identical questions.

3.3 Interpreting Drift: The Rashomon Effect

One theoretical account frames citation drift as a manifestation of the Rashomon effect, the well-documented tendency for multiple models, or a single model under stochastic decoding, to produce divergent but individually plausible outputs for the same underlying task (related work under the same research program, OpenReview submission 2frm73oO0t, 2025). Under this interpretation, citation instability is not simply an engineering defect to be patched away but a structural consequence of predictive multiplicity: many different citation sets can be locally coherent completions of a prompt, and a model sampling from its output distribution has no privileged mechanism for always selecting the same one. This framing has methodological implications for detection: rather than treating any deviation from a single 'ground truth' citation as an error, drift-aware evaluation should characterize the distribution of citations a model produces for equivalent prompts and assess whether that distribution is dominated by verifiable, relevant sources or by a diffuse mixture that includes fabricated or irrelevant ones.

4. Current Approaches to Detection

No single method currently addresses citation drift end to end. Instead, the emerging toolkit combines techniques originally developed for summarization faithfulness, hallucination detection, and bibliographic verification. We group current approaches into four families.

4.1 Retrieval-Grounded Citation Verification

The most direct approach to detecting citation problems is to retrieve the candidate publication from an external bibliographic source and compare its metadata against what the model produced. CiteCheck exemplifies this approach: given a candidate citation, the system retrieves plausible matching publications from external scholarly databases, compares the citation against each candidate using a structured LLM-based verifier, and classifies the result as exact, containing a minor metadata discrepancy, or containing a major discrepancy indicative of fabrication (Khajavi et al., 2026). On a purpose-built 982-citation physics benchmark with controlled corruptions ranging from subtle metadata drift to complete fabrication, this hybrid retrieval-plus-verification approach achieved 88.7 macro-F1 and 88.9% accuracy, outperforming zero-shot GPT, Claude, and Gemini baselines, including variants augmented with web search. This result suggests that purpose-built retrieval-grounded verification currently outperforms asking a general-purpose LLM to self-audit its own citations, a distinction with direct relevance for anyone deploying citation-checking as a safeguard layer on top of a summarization system.

4.2 Faithfulness Metrics Adapted from Summarization

A second family adapts metrics developed for abstractive summarization faithfulness to the citation-verification problem. NLI-based classifiers, in which the source passage is treated as a premise that must entail the generated claim, and question-answering-based consistency metrics, in which questions generated from the summary are answered against both the summary and the source and then compared, are the two most mature approaches in this category (Huang et al., 2025; Ji et al., 2023). These methods were not designed with structured bibliographic metadata in mind, but they are directly applicable to the content-fidelity half of the citation drift problem: verifying that the prose attached to a citation accurately represents the cited work, which is the concern raised by generalization-bias research (Peters & Chin-Yee, 2025). FactCC's joint prediction of sentence-level consistency and span-level evidence extraction (Kryscinski et al., 2020) offers a template for citation-aware faithfulness scoring, in which the supporting span would ideally be traceable not just to a source passage but to a specific, verifiable bibliographic entry.

4.3 Sampling- and Consistency-Based Detection

A third family avoids the need for an external reference corpus altogether by exploiting model self-consistency. SelfCheckGPT operationalizes the intuition that a model's stochastically sampled responses to the same prompt should agree on genuinely known facts but diverge on hallucinated ones, and demonstrates that inter-sample agreement, measured via BERTScore similarity, question-answering overlap, or NLI-based entailment across samples, correlates with human factuality judgments at both the sentence and passage level (Manakul et al., 2023). This family of methods maps naturally onto the citation drift entropy metric proposed by Ram (2025): both approaches operationalize reliability as the stability of outputs across repeated or resampled queries rather than as agreement with a fixed ground truth, making sampling-based consistency checking a particularly well-suited primitive for drift detection specifically, as distinct from one-shot fabrication detection.

4.4 LLM-as-Judge and Hybrid Pipelines

A fourth and increasingly common approach uses a separate, often more capable, LLM to evaluate the faithfulness or citation accuracy of a target model's output. Reviews of faithfulness evaluation across summarization, question answering, and machine translation consistently find that LLM-based judges correlate more closely with human evaluations than lexical or even embedding-based metrics, while cautioning that LLM judges are prone to false positives and can themselves hallucinate justifications for incorrect verdicts. In the citation-drift context specifically, CiteCheck's structured LLM verifier illustrates a hybrid design in which retrieval supplies grounding candidates and an LLM performs the fine-grained comparison, combining the precision of retrieval with the flexibility of learned judgment (Khajavi et al., 2026). Similarly, question-generation-and-answering pipelines built for related reproducibility problems, such as clinical hallucination detection frameworks that count and localize unsupported statements against a source document, illustrate how LLM-based judgment can be made more auditable by requiring the judge to point to specific supporting or contradicting evidence rather than issuing an unstructured verdict.

4.5 Summary of Detection Approaches

Approach	Core Mechanism	Representative Work
Retrieval-grounded verification	Retrieve candidate publication; compare metadata via structured LLM verifier	CiteCheck (Khajavi et al., 2026)
Adapted faithfulness metrics	NLI entailment / QA-based consistency between source and generated claim	FactCC (Kryscinski et al., 2020); surveys (Ji et al., 2023; Huang et al., 2025)

Approach	Core Mechanism	Representative Work
Sampling-based consistency	Inter-sample agreement across stochastic generations as an unreliability signal	SelfCheckGPT (Manakul et al., 2023); citation drift entropy (Ram, 2025)
LLM-as-judge / hybrid pipelines	A secondary LLM evaluates faithfulness or citation accuracy, often combined with retrieval	CiteCheck (Khajavi et al., 2026); RAG-grounded generation (Wang et al., 2025)

5. Research Challenges and Limitations

5.1 Conflation of Distinct Failure Modes

Much of the empirical literature reviewed in Section 2 reports a single aggregate 'fabrication rate' or 'accuracy rate' without distinguishing mutation, loss, fabrication, and content-level generalization bias as separate phenomena. This conflation makes it difficult to compare studies directly, since a paper reporting that '25% of citations are inaccurate' may be counting minor page-number errors and wholesale fabrications in the same statistic (Walters & Wilder, 2023; Mugaanyi et al., 2024). Citation drift research inherits this problem and compounds it with a temporal dimension: a reference that mutates from one turn to the next is a qualitatively different failure from one that is fabricated once and never mentioned again, yet both would register as 'unstable' under a coarse drift metric.

5.2 Absence of Standardized Multi-Turn Benchmarks

The detection methods reviewed in Section 4 were, with the partial exception of the citation drift study itself (Ram, 2025) and CiteCheck's controlled-corruption benchmark (Khajavi et al., 2026), developed and validated on single-response datasets. There is no widely adopted benchmark that systematically varies conversational depth, prompt paraphrasing, and topic domain while tracking the same underlying claim's citation across turns. The 240-conversation, 36-paper corpus introduced by Ram (2025) is a valuable first step but is limited to open-weight LLaMA models and six scientific domains, leaving open how proprietary, closed-weight systems and humanities or social-science literatures behave under the same protocol.

5.3 Model- versus Retrieval-Layer Attribution

For LLM systems that incorporate live retrieval, whether via RAG architectures (Lewis et al., 2020) or commercial web-search integration, it is often unclear whether observed citation instability originates in the language model's generation behavior or in fluctuations of the underlying retrieval index. Industry analyses of commercial answer engines have reported that the pool of sources cited for a fixed query can turn over substantially within days to weeks, a dynamic attributed to index refresh cycles and ranking changes rather than to model-level nondeterminism. While such analyses are not peer-reviewed academic studies and should be treated as preliminary observations rather than validated findings, the underlying concern is a legitimate methodological one: any evaluation of citation drift in retrieval-augmented systems needs to separate model-attributable drift from retrieval-attributable drift, and current academic benchmarks, which largely evaluate closed-book or single-snapshot-retrieval settings, do not yet do this systematically.

5.4 Reliance on LLM-Based Judges and Circularity Concerns

Several of the most accurate current detection systems, including CiteCheck's structured verifier (Khajavi et al., 2026) and general LLM-as-judge faithfulness evaluators, themselves rely on large language models to render verdicts. This introduces a potential circularity: a judge model trained or prompted similarly to the generator being evaluated may share its blind spots, and if both models draw on overlapping training corpora, systematic biases such as recency preference or the preferential citation of highly cited work documented by Algaba et al. (2025) could be reproduced rather than caught by the evaluator. Reviews of faithfulness evaluation methodology note that LLM judges, despite correlating best with human judgment among available automatic metrics, remain prone to false positives, a risk that is particularly consequential in a citation-verification context where a false negative (an undetected fabrication) could propagate into the scholarly record.

5.5 Language and Domain Coverage

Existing citation drift and fabrication studies are overwhelmingly conducted in English and concentrated in biomedicine, computer science, and general multidisciplinary topics. Multilingual faithfulness research on abstractive summarization more broadly has found that hallucination rates are higher, and existing faithfulness metrics less reliable, for non-English languages, since most faithfulness metrics and their underlying NLI or QA models were themselves developed and validated on English corpora. Whether citation drift specifically is more or less pronounced in non-English scholarly summarization, and whether the detection tools reviewed in Section 4 transfer to non-English bibliographic verification, remains largely unexamined.

6. Research Gaps and Future Directions

6.1 Longitudinal and Multi-Turn Evaluation Protocols

The single most direct extension suggested by this review is the development of standardized, publicly available benchmarks that track citation stability across paraphrased queries, follow-up requests, and extended conversations, spanning both open- and closed-weight models and a broader range of disciplines than the six domains covered by Ram (2025). Such benchmarks should report drift using disaggregated metrics, distinguishing mutation, loss, and fabrication rates rather than a single composite score, so that mitigation strategies targeting one mechanism can be evaluated without being obscured by the others.

6.2 Disentangling Model and Retrieval Contributions to Drift

For retrieval-augmented and web-connected summarization systems, future work should adopt controlled experimental designs that hold the retrieval index fixed while varying the language model, and vice versa, in order to isolate the respective contributions of parametric generation and non-parametric retrieval to observed citation instability. This would allow the field to move beyond the largely observational, industry-reported statistics currently available for commercial answer engines toward causally interpretable, peer-reviewed findings.

6.3 Citation-Aware Faithfulness Metrics

Existing summarization faithfulness metrics, including FactCC-style classifiers (Kryscinski et al., 2020) and the broader family of NLI- and QA-based methods (Huang et al., 2025), evaluate whether generated prose is supported by a source passage but were not designed to jointly verify that (a) a cited work exists, (b) its metadata is correctly reproduced, and (c) the prose attributed to it is an accurate representation of its actual content. Integrating retrieval-grounded citation verification, as in CiteCheck (Khajavi et al., 2026), with span-level faithfulness scoring in the style of FactCC represents a promising and largely unexplored architectural direction, potentially yielding a single pipeline that flags mutation, fabrication, and generalization bias within one unified evaluation.

6.4 Reducing Circularity in LLM-Based Verification

Given the circularity concerns raised in Section 5.4, future detection research should prioritize verifier architectures whose judgments are grounded in externally retrieved, independently verifiable evidence rather than relying purely on a second model's parametric judgment, and should report calibration statistics, particularly false-negative rates on fabricated citations, rather than aggregate accuracy alone, since the practical cost of a missed fabrication is substantially higher than that of a false alarm on a minor formatting discrepancy.

6.5 Extending Coverage Beyond English and STEM Fields

Given the near-total concentration of existing studies on English-language, STEM-adjacent literature, extending citation drift and fabrication research to humanities, social science, and non-English scholarly corpora is a clear and tractable gap. This is especially pressing because citation practices, publication norms, and the availability of open bibliographic metadata vary substantially across fields and languages, all of which plausibly affect both the rate of citation drift and the performance of retrieval-grounded verification tools that depend on the coverage of the bibliographic databases they query.

6.6 Standardized Reporting for AI-Assisted Scholarship

Finally, we note a broader infrastructural gap: there is currently no consensus mechanism by which publishers, preprint servers, or citation databases could record or flag AI-assisted summarization provenance, which would allow the kind of longitudinal drift tracking proposed above to be conducted at the scale of the published literature rather than solely within controlled laboratory benchmarks. Developing lightweight, interoperable standards for reporting when and how AI tools contributed to a citation list, paired with the technical detection methods reviewed in Section 4, would connect this line of research to ongoing editorial and policy discussions about generative AI in scientific publishing (Zhang & Zhao, 2025).

7. Conclusion

Citation drift names a problem that the literature on single-response citation fabrication had implicitly documented for several years without a unifying framework: that references generated by large language models are unstable not only in the sense that some fraction of them are simply invented, but in the sense that the same underlying claim can be cited differently, incompletely, or not at all depending on when and how it is asked about (Ram, 2025; Walters & Wilder, 2023). This review has argued that detecting citation drift in AI-generated literature summaries requires combining tools from three previously separate research threads: bibliographic fabrication auditing, summarization faithfulness evaluation, and retrieval-augmented generation. Retrieval-grounded verifiers such as CiteCheck currently offer the most accurate purpose-built detection of citation-level errors (Khajavi et al., 2026), while sampling-based consistency methods developed for general hallucination detection map naturally onto the specific problem of measuring drift across repeated queries (Manakul et al., 2023; Ram, 2025). At the same time, substantial gaps remain: standardized multi-turn benchmarks are scarce, the respective contributions of model-side and retrieval-side instability are rarely disentangled, LLM-based judges carry circularity risks that are not yet well characterized, and coverage of non-English and non-STEM scholarship is minimal. As AI-assisted summarization becomes further embedded in how researchers discover, digest, and cite the literature, closing these gaps is not merely a technical convenience but a matter of protecting the integrity of the scholarly record itself.

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